The **numspell** package

v1.5 (2021/11/23)

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1 Introduction

The aim of the `numspell` package is to spell the cardinal and ordinal numbers from 0 to $10^{66}-1$ (i.e. maximum 66 digits).

The supported languages are English (British and American), French, German, Hungarian, Italian, and Latin* (classical, medieval, modern, and ecclesiastic). The spelling will happen in the current language.

The `numspell` package requires the services of the `xstring` and `iflang` packages.

Load the package as usual, with

```
\usepackage{numspell}
```

2 Commands

`\numspell[`⟨*zeros*⟩`]{`⟨*num*⟩`}`
Spelling the cardinal number $n = \langle num\rangle \cdot 10^{\langle zeros\rangle}$, where $0 \le n \le 10^{66}-1$. The default value of ⟨*zeros*⟩ is `0`. For example

`\numspell{12000}` → twelve thousand
`\numspell[3]{12}` → twelve thousand
`\numspell[6]{12}` → twelve million
`\numspell[63]{1}` → one vigintillion

`\thenumspell`
The `\numspell` stores the result in this command. For example

`\numspell{12000}; \thenumspell` → twelve thousand; twelve thousand
`\numspell{1}; \numspell{2}; \thenumspell` → one; two; two

`\numspellsave{`⟨*name*⟩`}`
It generates the `\thenumspell`⟨*name*⟩ command, which saves the current `\thenumspell`. For example

*Thanks to Keno Wehr for the Latin language module.

```
\numspell{1};
\numspellsave{MyNum}
\numspell{2};
\thenumspell;
\thenumspellMyNum
```

one; two; two; one

`\numspelldashspace{`⟨*length*⟩`}`

In the number spelling, the spaces around the dashes are flexibility for the optimal hyphenation. Its value is `0pt plus` ⟨*length*⟩. The default value of ⟨*length*⟩ is `2pt`. For example

```
\selectlanguage{magyar}
\numspell{651231235476254716254625 4756}\\[2mm]
\numspelldashspace{10pt}
\numspell{651231235476254716254625 4756}
```

hatkvadrilliárd - ötszáztizenkétkvadrillió - háromszáztizenkéttrilliárd - háromszázötvennégytrillió - hétszázhatvankétbilliárd - ötszáznegyvenhétbillió - egyszázhatvankétmilliárd - ötszáznegyvenhatmillió-kétszázötvennégyezer-hétszázötvenhat

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`\numspell*[`⟨*zeros*⟩`]{`⟨*num*⟩`}`

It works like `\numspell`, but the number spelling will not be printed. In other words, the following two lines are equivalent:

```
\numspell[⟨zeros⟩]{⟨num⟩}
\numspell*[⟨zeros⟩]{⟨num⟩}\thenumspell
```

For example

```
\numspell*{1}
\numspellsave{MyNum}
\numspell*{2}
\thenumspell;
\thenumspellMyNum
```

two; one

`\Numspell[`⟨*zeros*⟩`]{`⟨*num*⟩`}`

It works like `\numspell`, but the first letter will be capital. For example

`\Numspell{12000}` → Twelve thousand
`\Numspell[3]{12}` → Twelve thousand
`\Numspell[6]{12}` → Twelve million
`\Numspell[63]{1}` → One vigintillion

`\Numspell*[`⟨*zeros*⟩`]{`⟨*num*⟩`}`

It works like `\Numspell`, but the number spelling will not be printed. In other words, the following two lines are equivalent:

`\Numspell[`⟨*zeros*⟩`]{`⟨*num*⟩`}`
`\Numspell*[`⟨*zeros*⟩`]{`⟨*num*⟩`}\thenumspell`

For example

```
\Numspell*{1}
\numspellsave{MyNum}
\Numspell*{2}
\thenumspell;
\thenumspellMyNum
```

Two; One

`\ordnumspell[`⟨*zeros*⟩`]{`⟨*num*⟩`}`
Spelling the ordinal number $n = \langle num \rangle \cdot 10^{\langle zeros \rangle}$, where $0 \leq n \leq 10^{66} - 1$. The default value of ⟨*zeros*⟩ is `0`. For example

`\ordnumspell{12000}` → twelve thousandth
`\ordnumspell[3]{12}` → twelve thousandth
`\ordnumspell[6]{12}` → twelve millionth
`\ordnumspell[63]{1}` → one vigintillionth

`\ordnumspell*[`⟨*zeros*⟩`]{`⟨*num*⟩`}`
It works like `\ordnumspell`, but the number spelling will not be printed. In other words, the following two lines are equivalent:

`\ordnumspell[`⟨*zeros*⟩`]{`⟨*num*⟩`}`
`\ordnumspell*[`⟨*zeros*⟩`]{`⟨*num*⟩`}\thenumspell`

For example

```
\ordnumspell*{1}
\numspellsave{MyNum}
\ordnumspell*{2}
\thenumspell;
\thenumspellMyNum
```

second; first

`\Ordnumspell[`⟨*zeros*⟩`]{`⟨*num*⟩`}`
It works like `\ordnumspell`, but the first letter will be capital. For example

`\Ordnumspell{12000}` → Twelve thousandth
`\Ordnumspell[3]{12}` → Twelve thousandth
`\Ordnumspell[6]{12}` → Twelve millionth
`\Ordnumspell[63]{1}` → One vigintillionth

`\Ordnumspell*[`⟨*zeros*⟩`]{`⟨*num*⟩`}`
It works like `\Ordnumspell`, but the number spelling will not be printed. In other words, the following two lines are equivalent:

`\Ordnumspell[`⟨*zeros*⟩`]{`⟨*num*⟩`}`
`\Ordnumspell*[`⟨*zeros*⟩`]{`⟨*num*⟩`}\thenumspell`

For example

```
\Ordnumspell*{1}
\numspellsave{MyNum}
\Ordnumspell*{2}
\thenumspell;
\thenumspellMyNum
```

Second; First

3 Commands for English language

If the `english`, `british`, `ukenglish` or `UKenglish` language is active, then the number spelling will happen in British English. But it will be in American English, if the `american`, `usenglish` or `USenglish` language is active.

`\numspellUS`
Using British English, you can rechange the number spelling to American English by this command.

`\numspellGB`
Using American English, you can rechange the number spelling to British English by this command.

4 Commands for French language

The following commands only work, if `french` language is active.

`\numspellpremiere`
By default, `\ordnumspell{1}` → premier,
but `\numspellpremiere\ordnumspell{1}` → première

`\numspellpremier` (default)
`\numspellpremiere\ordnumspell{1};`
`\numspellpremier\ordnumspell{1}`

première ; premier

5 Commands for Hungarian language

The following commands only work, if `magyar` or `hungarian` language is active.

\anumspell[⟨zeros⟩]{⟨num⟩}

It works like \numspell, but the number spelling will start with Hungarian definite article. For example

\anumspell{1} → az egy
\anumspell{2} → a kettő

\anumspell*[⟨zeros⟩]{⟨num⟩}

It works like \anumspell, but the number spelling will not be printed. In other words, the following two lines are equivalent:

\anumspell[⟨zeros⟩]{⟨num⟩}
\anumspell*[⟨zeros⟩]{⟨num⟩}\thenumspell

For example

```
\anumspell*{1}
\numspellsave{MyNum}
\anumspell*{2}
\thenumspell;
\thenumspellMyNum
```

a kettő; az egy

\Anumspell[⟨zeros⟩]{⟨num⟩}

It works like \anumspell, but the first letter will be capital.

\Anumspell*[⟨zeros⟩]{⟨num⟩}

It works like \anumspell*, but the first letter will be capital.

\aordnumspell[⟨zeros⟩]{⟨num⟩}

It works like \ordnumspell, but the number spelling will start with Hungarian definite article. For example

\aordnumspell{1} → az első
\aordnumspell{2} → a második

\aordnumspell*[⟨zeros⟩]{⟨num⟩}

It works like \aordnumspell, but the number spelling will not be printed. In other words, the following two lines are equivalent:

\aordnumspell[⟨zeros⟩]{⟨num⟩}
\aordnumspell*[⟨zeros⟩]{⟨num⟩}\thenumspell

For example

```
\aordnumspell*{1}
\numspellsave{MyNum}
\aordnumspell*{2}
\thenumspell;
\thenumspellMyNum
```

a második; az első

\Aordnumspell[⟨zeros⟩]{⟨num⟩}
It works like \aordnumspell, but the first letter will be capital.

\Aordnumspell*[⟨zeros⟩]{⟨num⟩}
It works like \aordnumspell*, but the first letter will be capital.

6 Commands for Italian language

The following commands only work, if italian language is active.

\numspellitmasculine (default)
The ordinal numbers will be printed in masculine form. For example
\ordnumspell{1} → primo

\numspellitfeminine
The ordinal numbers will be printed in feminine form. For example
\numspellitfeminine\ordnumspell{1};
\numspellitmasculine\ordnumspell{1}

prima; primo

7 Commands for the Latin language

The following commands only work, if one of the next languages is active: latin, classiclatin, medievallatin, or ecclesiasticlatin.

\numspelllamasculine (default)
All numbers will be printed in masculine form. For example
\numspell{1}, \numspell{2}, \numspell{200}, \ordnumspell{1}

unus, duo, ducenti, primus

\numspelllafeminine
All numbers will be printed in feminine form. For example
\numspelllafeminine
\numspell{1}, \numspell{2}, \numspell{200}, \ordnumspell{1}

una, duae, ducentae, prima

\numspelllaneuter
All numbers will be printed in neuter form. For example
\numspelllaneuter
\numspell{1}, \numspell{2}, \numspell{200}, \ordnumspell{1}

unum, duo, ducenta, primum

8 Examples

```
\documentclass{article}
\usepackage[T1]{fontenc}
\usepackage[magyar,italian,latin,ngerman,french,english]{babel}
\usepackage{numspell}
\usepackage[group-separator={,}]{siunitx}
\begin{document}
\def\mynum{123456789}
\noindent
In British English the spelling of \num{\mynum} is
\emph{\numspell{\mynum}}.
\par\smallskip\noindent
In American English the spelling of \num{\mynum} is
\foreignlanguage{american}{\em\numspell{\mynum}}.
\par\smallskip\noindent
In French the spelling of \num{\mynum} is
\foreignlanguage{french}{\em\numspell{\mynum}}.
\par\smallskip\noindent
In German the spelling of \num{\mynum} is
\foreignlanguage{ngerman}{\em\numspell{\mynum}}.
\par\smallskip\noindent
In Hungarian the spelling of \num{\mynum} is
\foreignlanguage{magyar}{\em\numspell{\mynum}}.
\par\smallskip\noindent
In Italian the spelling of \num{\mynum} is
\foreignlanguage{italian}{\em\numspell{\mynum}}.
\par\smallskip\noindent
In Latin the spelling of \num{\mynum} is
\foreignlanguage{latin}{\em\numspell{\mynum}}.
\end{document}
```

In British English the spelling of 123,456,789 is *one hundred and twenty-three million, four hundred and fifty-six thousand, seven hundred and eighty-nine.*

In American English the spelling of 123,456,789 is *one hundred twenty-three million, four hundred fifty-six thousand, seven hundred eighty-nine.*

In French the spelling of 123,456,789 is *cent vingt-trois millions quatre cent cinquante-six mille sept cent quatre-vingt-neuf.*

In German the spelling of 123,456,789 is *einhundertdreiundzwanzig Millionen vierhundertsechsundfünfzigtausendsiebenhundertneunundachtzig.*

In Hungarian the spelling of 123,456,789 is *százhuszonhárommillió-négyszázötvenhatezer-hétszáznyolcvankilenc.*

In Italian the spelling of 123,456,789 is *centoventitré milioni quattrocentocinquantaseimila settecentottantanove.*

In Latin the spelling of 123,456,789 is *centum viginti tres milliones quadringenta quinquaginta sex milia septingenti undenonaginta.*

```
\documentclass{article}
\usepackage{numspell}
\newcounter{mycount}
\begin{document}
The
\makeatletter
\@whilenum\value{mycount}<51
\do{\ordnumspell{\themycount}\stepcounter{mycount},\ }\dots
\makeatother
\end{document}
```

The zeroth, first, second, third, fourth, fifth, sixth, seventh, eighth, ninth, tenth, eleventh, twelfth, thirteenth, fourteenth, fifteenth, sixteenth, seventeenth, eighteenth, nineteenth, twentieth, twenty-first, twenty-second, twenty-third, twenty-fourth, twenty-fifth, twenty-sixth, twenty-seventh, twenty-eighth, twenty-ninth, thirtieth, thirty-first, thirty-second, thirty-third, thirty-fourth, thirty-fifth, thirty-sixth, thirty-seventh, thirty-eighth, thirty-ninth, fortieth, forty-first, forty-second, forty-third, forty-fourth, forty-fifth, forty-sixth, forty-seventh, forty-eighth, forty-ninth, fiftieth, ...

```
\documentclass{article}
\usepackage{numspell}
\newcounter{mycount}
\def\themycount{\numspell{\arabic{mycount}}}
\begin{document}
\Numspell{0},
\makeatletter
\@whilenum\value{mycount}<30
\do{\stepcounter{mycount}\themycount,\ }\dots
\makeatother
\end{document}
```

Nought, one, two, three, four, five, six, seven, eight, nine, ten, eleven, twelve, thirteen, fourteen, fifteen, sixteen, seventeen, eighteen, nineteen, twenty, twenty-one, twenty-two, twenty-three, twenty-four, twenty-five, twenty-six, twenty-seven, twenty-eight, twenty-nine, thirty, ...

9 Limitations

Do not use the `\numspell`, `\numspell*`, `\Numspell`, `\Numspell*`, etc. commands inside sectioning commands and `\MakeUppercase`. An example for the illustration of the problem:

```
\documentclass{article}
\usepackage{hyperref}
\usepackage{numspell}
\pagestyle{headings}
```

```
\begin{document}
\section{The \ordnumspell{123} factor}
\MakeUppercase{\numspell{123}}
\newpage
Text
\end{document}
```

The following mistakes occur:

1. On the page 1: “one hundred and twenty-three”
   Required: “ONE HUNDRED AND TWENTY-THREE”
2. On the heading: *“THE one hundred and twenty-third FACTOR”*
   Required: *“THE ONE HUNDRED AND TWENTY-THIRD FACTOR”*
3. On the pdf bookmark: “The 123 factor”
   Required: “The one hundred and twenty-third factor”

The solution

```
\documentclass{article}
\usepackage{hyperref}
\usepackage{numspell}
\pagestyle{headings}
\begin{document}
\ordnumspell*{123}
\section{The \thenumspell\ factor}
\numspell*{123}
\MakeUppercase{\thenumspell}
\newpage
Text
\end{document}
```